

TrES-3b

R-1490

Benchmark run · workflow W8-benchmark-deep · record deep-bench_TRES-3_190510 · created 2026-08-01T13:45:17+00:00

BENCHMARK: NON-DETECTION

Results

Mid-Transit Time (Tmid)	2458613.8847 +/- 0.0014 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.163 +/- 0.057
Transit depth (Rp/Rs)^2	2.67 +/- 1.85 [%]
Orbital Inclination (inc)	86.72 +/- 2.89
Airmass coefficient 1 (a1)	0.045 +/- 0.026
Airmass coefficient 2 (a2)	3.0 +/- 1.56
Scatter in the residuals of the lightcurve fit is	58.92 %
Best Comparison Star	None
Optimal Aperture	3.0
Optimal Annulus	3.0
Transit Duration (day)	0.077 +/- 0.012

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R $\blacksquare$ fit vs published	0.163 $\pm$ 0.057 vs 0.16309 (deviation 0.0 σ)
Duration fit vs published	0.077 d vs 0.0567 d (deviation 1.69 σ)
Mid-time O–C	2.2 min
Depth SNR	2.9
Residual scatter	58.92 %

Light curve

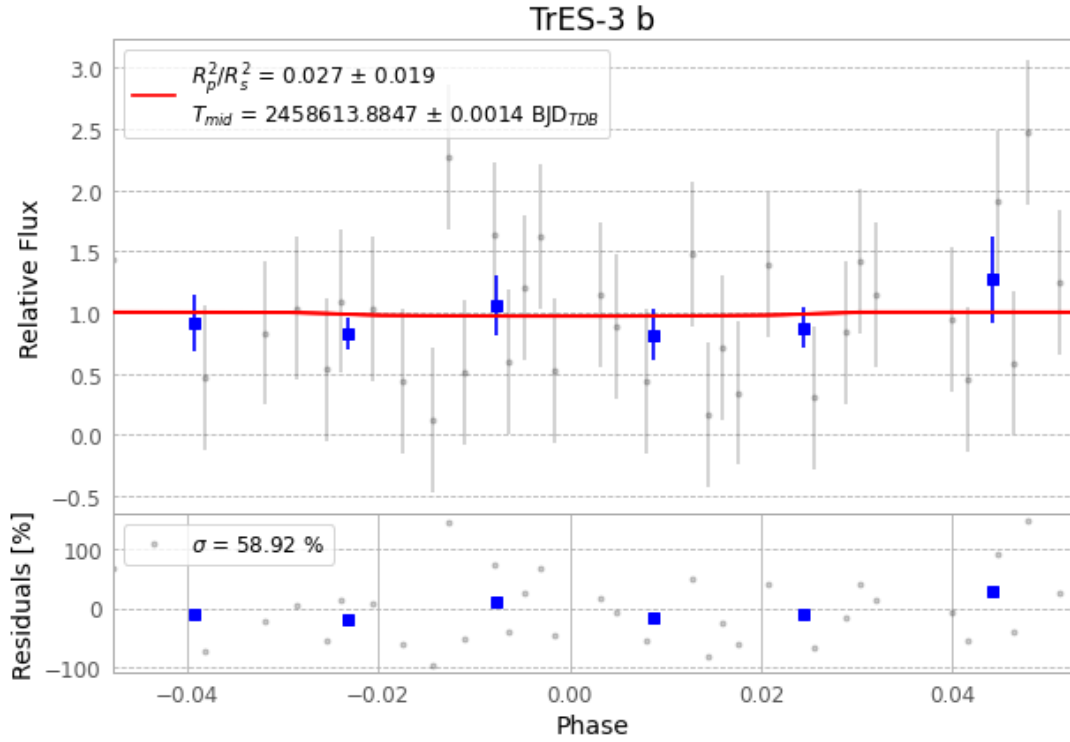


Figure 1 — FinalLightCurve_TrES-3 b_2019-05-10.png (SHA-256 f0f76416b5cfef9a...)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [590, 372], 3 comparison star(s). Exact configuration: parameters.inits_json in record.json (SHA-256 1eba8e9b6ea97f5c...).

exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline_commit ba3ef8b

Data provenance

66 FITS frames (43 MB), TRES-3190510073312.FITS → TRES-3190510104812.FITS. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-TRES-3-190510.log (0fb2c3a4191d...), FinalLightCurve_TrES-3 b_2019-05-10.png (f0f76416b5cf...), FinalLightCurve_TrES-3 b_2019-05-10.csv (45bfb6e65dcf4...)

Compute resources

Wall time 145.3 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)

Observer: — · AAVSO obscode ABGA · Automated pipeline with human review — nothing is submitted without a human approving this specific result. Data license CC-BY-4.0. Generated 2026-08-01 13:45Z.